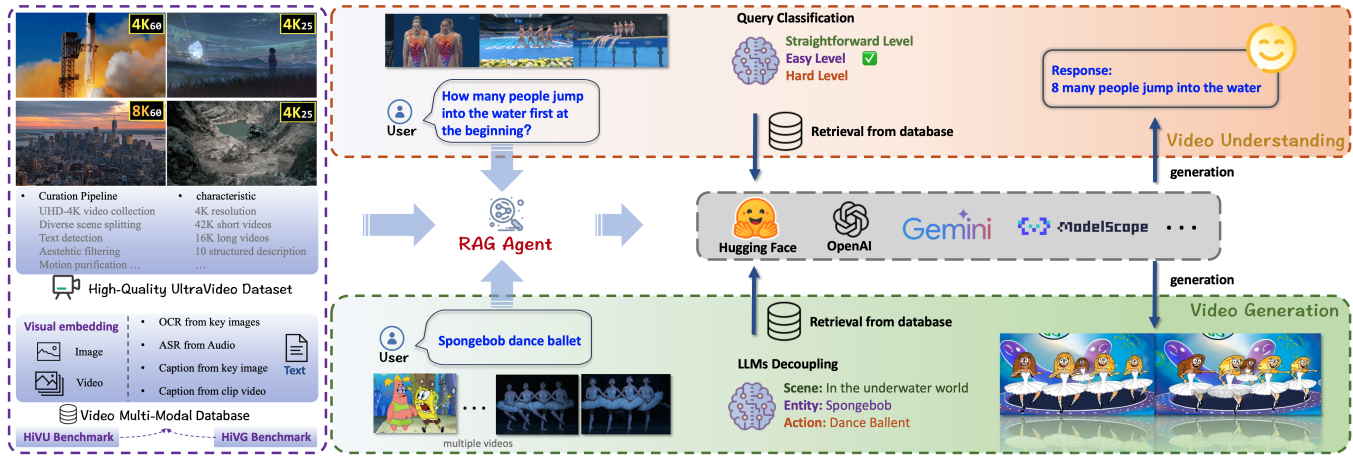


# Multi-Modal Retrieval Augmented Visual Understanding and Generation

Zhucun Xue  
Zhejiang University  
Zhejiang, China  
12432038@zju.edu.cn



**Figure 1: Research trajectory of the doctoral symposium paper.** I conduct research on visual understanding and generation with Retrieval-Augmented Generation (RAG) as the core. Specifically, I start from retrieval-augmented video understanding (Sec. 2) and generation (Sec. 3), and plan to extend it to agent RAG (Sec. 4) with more application value in the future. I have also proposed the high-resolution and high-quality UltraVideo (Sec. 5) dataset to support the research. In addition, RAG-based visual task-specific benchmarks are proposed to evaluate the proposed methods.

## Abstract

This paper presents a doctoral research focusing on integrating Retrieval-Augmented Generation (RAG) into video-related multi-modal tasks. Existing RAG studies predominantly target text, images, or tabular data, overlooking the unique value of video as a knowledge carrier. We address this gap by: 1) proposing AdaVideoRAG, a framework that adaptively allocates retrieval strategies based on query complexity for long-video understanding; 2) developing REVIG (RAG-Enhanced Video Generation) to optimize prompt engineering via retrieved knowledge for controllable video synthesis; 3) constructing the UltraVideo dataset (UHD-4K/8K resolution, 100+ themes, 10 structured captions per video) and HiVU/HiVG benchmarks to evaluate RAG-driven video tasks. Experiments validate the effectiveness of our methods, and we outline future plans to unify video understanding and generation through Agentic RAG for AGI-oriented research.

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## CCS Concepts

• Computing methodologies → Computer vision.

## Keywords

Retrieval Augmented Generation, Visual Understanding and Generation, Dataset and Benchmark

## ACM Reference Format:

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## 1 Introduction

Retrieval-Augmented Generation (RAG) dynamically integrates external knowledge to resolve LLMs' knowledge solidification and hallucinations [7, 12, 13]. Its built-in retrieval leverages beyond-training data sources, enabling domain-specific personalization without LLM fine-tuning [8, 35], while maintaining knowledge timeliness and verifiability.

Current RAG research mostly focuses on text modality [7, 9, 12], static images [6], and tabular forms [4], overlooking the unique value of video as a multimodal knowledge carrier. Especially in tasks such as long video understanding, dynamic scene generation, and interactive video QA, *how to use RAG to enhance the capabilities of Multimodal Large Language Models (MLLMs) is a*

*question worthy of exploration, and it is also the topic mainly discussed and studied in this paper.* Specifically, RAG can extract multi-source information from videos, including audio, caption, and OCR, which can retrieve direct video segments related to the query as well as text descriptions, helping us to better understand the video comprehensively and accurately. Especially in long video understanding, due to the current limitations of MLLMs on the input video length and text tokens, using RAG can retrieve the most relevant information from long videos, thus avoiding the loss of valid information and the interference of noise. On the other hand, video generation is also a promising task that holds significant value in fields such as film and television production scenarios and world model research. It complements the RAG understanding model and can also benefit from it. As a doctoral research topic worthy of in-depth exploration, I will gradually introduce the completed and planned research (Fig. 1) from the following aspects.

**1. RAG Understanding (Sec. 2).** Currently, most of the RAG research on long video attempts to leverage LLMs and domain-specific tools to acquire multi-modal information for creating a retrieval database [1, 16, 21, 22, 34]. Among them, [16] adopts the vector retrieval approach to obtain the text information most relevant to the query. However, this kind of method exhibits insufficient capabilities when handling sensemaking queries or multi-hop questions. VideoRAG [21], by constructing a graph-retrieval strategy, significantly enhances the logical association of long-video context information. Nevertheless, it needs to maintain a hierarchical graph database, which demands substantial computational and time resources. Moreover, it incurs higher costs when migrating to new scenarios. Therefore, rationally allocating queries of different videos and difficulty levels and how to use RAG to efficiently utilize resources are among the main issues pondered in this article.

**2. RAG Generation (Sec. 3).** RAG can help retrieve relevant videos or the required motion templates in the video generation task, assisting in generating high-quality, coherent, and contextually accurate videos. Recent methods [10, 20, 28] will perform supervised training or fine-tuning on the Text-to-Video(T2V) model using videos retrieved from short videos that contain only the information of one action sequence. However, this does not meet our original intention: based on existing MLLMs, without post-training, we can use plug-and-play RAG to assist MLLMs obtaining more reliable results. Notably, early research on RAG applications in video generation is relatively scarce, mainly due to urgent technical challenges in its complex retrieval mechanism (needing to capture video spatio-temporal continuity and multimodal information) and model adaptation mechanism (accurately integrating retrieval results to guide dynamic content generation)—issues also studied in this paper.

**3. RAG Agent (Sec. 4).** With the popularity of Artificial General Intelligence(AGI) [18], Agentic RAG [23, 29], by integrating the autonomous decision-making ability of agents and the dynamic knowledge enhancement mechanism of multimodal RAG, provides a disruptive solution for video tasks. Through dynamic task decomposition and multi-agents collaboration, Agentic RAG can more flexibly and actively schedule external tools and iteratively optimize. It may be the future solution direction that can combine video understanding and generation, promoting video AI from the "passive" to the "active thinking" intelligent era.

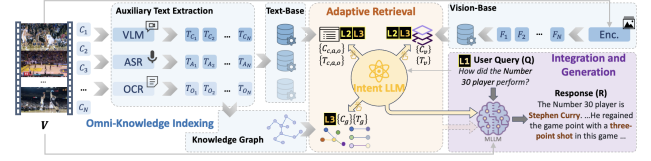


Figure 2: Overview of our AdaVideoRAG framework.

**4. Video Dataset and Benchmark (Sec. 5).** To better evaluate RAG’s effectiveness in video understanding and generation, dataset and benchmark construction is crucial. First, text-to-video (T2V) model performance is heavily constrained by training data quality—especially in visual resolution, temporal consistency, and fine-grained text-semantic alignment. While existing large-scale T2V datasets [14, 19, 26] are quantity-abundant, they focus on medium/low-resolution content (e.g., 720p) and simple subtitles [3, 5, 11, 17, 24, 26, 27, 30], failing to meet Ultra-High Definition (UHD) video requirements (sharp details, rich textures, fine semantic control). Second, long-video understanding lacks high-quality benchmarks with sufficient quantity, length, and categories; retrieval-augmented video generation research, still in infancy, has no suitable evaluation benchmark.

In response to the above analysis, my research contributions are as follows: **1)** We propose a novel AdaVideoRAG framework [32] that adaptively allocates appropriate retrieval strategies based on query complexity. **2)** We present the RAG-Enhanced Video Generation (REViG) framework, which leverages retrieved information to optimize prompt engineering for controllable video generation. **3)** We introduce a high-quality open-sourced UHD-4K text-to-video dataset named UltraVideo [31], covering over 100 diverse themes, with each video accompanied by 10 structured captions. Additionally, we establish the Hierarchical Video Understanding (HiVU) and Hierarchical Video Generation (HiVG) benchmarks for video understanding and generation tasks. **4)** Experiments validate the effectiveness of our proposed methods, and we plan to explore a new direction of unifying video understanding and generation using agents in the future.

## 2 Retrieval-Augmented Video Understanding

To address the dual challenges of low efficiency and insufficient accuracy in unified retrieval mechanisms caused by the differences in user query complexity, we propose an adaptive RAG framework termed AdaVideoRAG (Fig. 2), which leverages LLMs to classify query intent and dynamically switches between lightweight retrieval (single-frame features) and deep retrieval (cross-segment temporal logic and multimodal context associations), improving accuracy for complex queries while ensuring response efficiency.

### 2.1 Level-1: Straightforward Reasoning.

For straightforward reasoning queries (e.g., object color recognition), we activate a lightweight channel: the query  $Q$  and full video  $\{C_n\}$  are routed directly to the MLLM. This leverages the MLLM’s intrinsic knowledge/reasoning capabilities while bypassing external knowledge retrieval, eliminating resource overhead from complex cross-modal alignment.

## 2.2 Level-2: Simple Reasoning.

In single-step reasoning tasks (e.g., "why the girl cried before rain"), existing MLLMs' coarse cross-modal integration breaks key evidence chains, which ignores audio emotions or subtitle warnings, thus failing to associate causal motivations.

For this Simple reasoning, we divide the video into  $N$  consecutive and semantically complete clips  $V = (C_1, C_2, \dots, C_N) = \{C_n\}$  according to a fixed duration (30s per clip in the paper). Then we extract the auxiliary text information  $T_{C,A,O}$  contained in each clip to create a text database  $D_{C,A,O}$ , including video captions, ASR from audio, OCR from key frames, and also create a vision database  $D_V$  which is the concatenated visual embeddings.

During the retrieval process, the original query is first decoupled by LLMs into modality-specific sub-queries. These sub-queries then compute similarity scores against: the caption database from video clips  $D_C$ , the ASR database from audio clips  $D_A$ , and the OCR database from key frames  $D_O$ , respectively. This retrieves the top-k relevant vectors for each modality. Based on index mapping, corresponding original video clips  $C_{c,a,o}$  and auxiliary text  $T_{c,a,o}$  are then located. Furthermore, we reuse the rewritten captions as semantic anchors to map video clips into the text embedding space  $\{F_n\}$  via the ImageBind encoder [], obtaining the vision database  $D_V$ . We calculate the cosine similarity between text-visual embeddings, filter candidate segments exceeding the 0.5, and retain the highest-confidence Top-K visual evidence  $C_v$ . Finally, visual evidence  $C_v$  is merged and filtered with auxiliary text-localized video evidence  $C_{c,a,o}$  to construct the Level-2 simple reasoning retrieval results.

Finally, we use the retrieved results and send them together with the query  $Q$  into any MLLMs, which can achieve a more comprehensive and fine-grained understanding of the basic model on long videos as  $R = \text{MLLM}(\{C_v\}, \{C_{c,a,o}\}, \{T_{c,a,o}\}, Q)$ .

## 2.3 Level-3: Hard Reasoning.

For Hard reasoning, this highest difficulty video understanding task requires extracting multi-entity associations from contexts, constructing an entity-relationship knowledge graph across spatio-temporal and semantic dimensions, and integrating the deep reasoning capabilities of MLLMs to be achieved. For example, for the question "What life lessons does the movie contain?", the system must excavate the implicit semantic associations in the video and perform multi-hop reasoning to generate accurate answers.

Therefore, we specifically create a retrieval method with a higher complexity for queries at Level-3. We utilize all the multimodal information  $T_{C,A,O}$  extracted from the original video, extract the entities and relationships contained therein, and use them to create a knowledge graph  $G$ . After that, we reuse the decoupled caption query to perform similarity matching on the vectors in  $G$ , and retrieve the query-centered sub-graph  $g$ . At the same time, we also retrieve the visual clips  $C_v$  that are most relevant to the query scene description from the vision database  $D_V$ , perform merging and filtering, and obtain the Level3 retrieval results as  $R = \text{MLLM}(\{C_v\}, \{g\}, \{T_{c,a,o}\}, Q)$ .

## 3 Retrieval-Augmented Video Generation

In the video generation task, we use RAG to help us retrieve the content needed for generation. Some previous works focused more

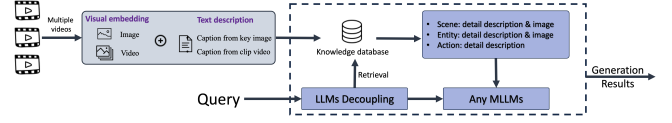


Figure 3: Overview of our REVIG framework.

on using video captions for retrieval in short video clips. This paper places more emphasis on finding video clips related to query from multiple long videos and proposes a novel RAG-Enhanced Video Generation (REVIG) framework (Fig. 3), making it play a role in the Text-to-Video (T2V) and Image-to-Video (I2V) generation tasks.

**I. Creation of a Multi-Video Database.** We build a modality database by splitting multiple videos: divide each video into a sequence of clips, and perform four steps of processing for each clip: 1) Generate video clip captions; 2) Extract key frames to generate captions; 3) Encode key frame features through ImageBind; 4) Concatenate all the key frame features within the clip. Finally, we construct four core databases: the video clip caption database  $V_C$ , the key frame caption database  $I_C$ , the key frame feature embedding database  $I_V$ , and the video feature embedding database  $V_V$ .

**II. Multi-Modal Retrieval and Generation.** In the video generation task, the system decouples the user's Query into three elements through LLMs: Scene (scene description), Entity (entity object), Action (dynamic behavior), and conducts specialized retrieval for each element respectively to achieve goal-oriented generation control.

Among them, for the retrieval of Scene and Entity, only a single image or its description is required to meet the information needs. Therefore, the image database  $I_C$  and the image encoding library ( $I_V$ ) are retrieved in parallel to obtain the most relevant image description  $I_c$  and image encoding  $I_v$ . After reranking, the image id corresponding to the result with the highest confidence is selected as the controllable output frame, and its associated image caption is used as the controllable text prompt:  $T_{scene}$ ,  $T_{entity}$ . For the generation of the dynamic behavior of Action, it is necessary to analyze the temporal and spatial information. Therefore, based on the joint retrieval results of the video clip caption database  $V_C$  and the video feature embedding database  $V_V$ , a text prompt  $T_{action}$  for temporal action control is generated.

The final generation can be represented by the following formula:

$$R = \begin{cases} \text{MLLM}(T_{scene}, T_{entity}, T_{action}, Q) & \text{in T2V task,} \\ \text{MLLM}(I, T_{scene}, T_{entity}, T_{action}, Q) & \text{in I2V task.} \end{cases} \quad (1)$$

The controllable input  $I$  can be selected as the image corresponding to the retrieved scene or entity.

## 4 Retrieval-Augmented Agent (Future Work)

By introducing the Agent architecture, we can build a video processing hub that can intelligently schedule external tools, seamlessly integrating multi-modal video understanding and generation tasks into a unified engineering workflow. At the same time, the Agent can also autonomously decide the tool chain according to the task complexity. For simple queries (such as entity localization), it triggers a lightweight vision API. For complex requirements (such as causal event generation), it collaborates with a knowledge retriever, a spatio-temporal reasoning engine, and a generation model to achieve a closed-loop workflow of "understanding - planning - generation", reducing the system coupling degree while improving

**Table 1: Results on MLVU\_test [36] dataset.**

| Model                     | Params | Frames   | TR   | AR   | NQA  | ER   | PQA  | SQA  | AO   | AC   | TQA  | AVG  | Gain  |
|---------------------------|--------|----------|------|------|------|------|------|------|------|------|------|------|-------|
| GPT-4o                    | -      | 0.5fps   | 83.7 | 68.8 | 42.9 | 47.8 | 57.1 | 63.6 | 46.2 | 35   | 48.7 | 54.9 | -     |
| Video-LLaVA               | 7B     | 8        | 64.4 | 35.9 | 25.4 | 34   | 26   | 25   | 13.1 | 16.9 | 23.8 | 29.4 | -     |
| Video-LLaVA + AdaVideoRAG | 7B     | 8        | 73.9 | 33.1 | 46.2 | 38   | 41.9 | 31.3 | 21.2 | 16.9 | 38.5 | 37.9 | 28.9% |
| Qwen2.5-VL                | 7B     | 2fps-768 | 46.7 | 15.4 | 16.9 | 35.8 | 38   | 38.9 | 24.6 | 13.6 | 31   | 29.0 | -     |
| Qwen2.5-VL + AdaVideoRAG  | 7B     | 2fps-768 | 78.9 | 30.8 | 44.1 | 37.7 | 48   | 36.1 | 33.3 | 15.3 | 40.5 | 40.5 | 39.8% |
| Qwen2.5-VL                | 72B    | 2fps-768 | 73.3 | 33.3 | 59.3 | 47.2 | 40   | 41.7 | 37.7 | 16.9 | 26.2 | 41.7 | -     |
| Qwen2.5-VL + AdaVideoRAG  | 72B    | 2fps-768 | 82.2 | 41   | 54.2 | 41.5 | 44   | 47.2 | 35.1 | 15.1 | 45.2 | 45.1 | 8%    |
| VideoLLaMA3               | 7B     | 1fps-180 | 76.9 | 43.6 | 68.3 | 54.7 | 58   | 34.3 | 25   | 33.3 | 34.9 | 47.7 | -     |
| VideoLLaMA3 + AdaVideoRAG | 7B     | 1fps-180 | 83.8 | 47.1 | 69.2 | 62.3 | 64   | 38.9 | 34.8 | 35.6 | 42.9 | 53.2 | 11.6% |

**Table 2: Performance on HiVU benchmark.**

| Metric            | Level-2     |                            | Level-3     |                            | Overall     |                            |
|-------------------|-------------|----------------------------|-------------|----------------------------|-------------|----------------------------|
|                   | VideoLLaMA3 | VideoLLaMA3 w/ AdaVideoRAG | VideoLLaMA3 | VideoLLaMA3 w/ AdaVideoRAG | VideoLLaMA3 | VideoLLaMA3 w/ AdaVideoRAG |
| Comprehensiveness | 42.72%      | 57.28%                     | 26.00%      | 74%                        | 35.98%      | 64.02%                     |
| Empowerment       | 36.81%      | 63.19%                     | 25.11%      | 74.89%                     | 30.88%      | 69.12%                     |
| Trustworthiness   | 36.81%      | 63.19%                     | 26.45%      | 73.55%                     | 30.58%      | 69.42%                     |
| Depth             | 34.09%      | 65.91%                     | 22.87%      | 77.13%                     | 26.23%      | 73.77%                     |
| Density           | 38.63%      | 61.37%                     | 25.11%      | 74.89%                     | 31.03%      | 68.97%                     |
| Overall Winner    | 37.27%      | 62.73%                     | 22.87%      | 77.13%                     | 30.58%      | 69.42%                     |

the cross-task collaboration efficiency. This is also the research direction of our future work.

## 5 Video Dataset and Benchmark

### High-Quality UltraVideo Dataset.

This paper first proposes a high-quality open-sourced UHD-4K (with 30% 8K) text-to-video dataset named UltraVideo [31] as shown in Fig. 1, which covers extensive themes (over 100 categories), with each video accompanied by 10 structured captions (averaging 824 words). Specifically, we meticulously design a 4-stage highly automated curation process to obtain the final high-quality dataset: *i*) Diverse and high-quality video clip collection from various scenes. *ii*) Statistical data filtering including text, blackboard, exposure, and graying detections. *iii*) Model-based data purification involving aesthetic and temporal motion scoring, video-caption consistency, and MLLM-assisted attribute judgment. *iv*) Comprehensive structured caption generation. We further extend WAN to UltraWAN-4K to natively generates high-quality 4K videos with more consistent text controllability, demonstrating the effectiveness of our data curation. **HiVU and HiVG Benchmarks.** We introduce the first holistic video understanding benchmark (HiVU): (1) 120 videos spanning 1-106min across knowledge, news, and entertainment genres; (2) quantifiable three-level queries (L1 frame-perception/L2 temporal-reasoning/L3 cross-modal inference)

We also release a Hierarchical Video Generation (HiVG) benchmark, which includes videos of different types and durations. During video generation, models need locate relevant video segments, posing challenges to both retrieval quality and generation ability.

## 6 Experimental Results

The paper only presents main experiments due to space constraints.

**Videos Understanding of AdaVideoRAG.** As shown in Table Tab. 1, in the MLVU test set, the adaptiveRAG based strategy proposed in this paper significantly improves the accuracy of all MLLMs in video understanding, including the baseline model Video: LLaVA [15] and the open-source models Qwen2.5-VL series [2], VideoLLaMA3 [33]. The improvement is particularly prominent in multi-step reasoning tasks (such as Topic Reasoning) and global reasoning tasks (such as Action Counting). Empirical results show that: 1) After Qwen2.5-VL-7B adopts AdaVideoRAG, its accuracy increases by 40 percent, and its performance is on a par with its 32B model; 2) The lightweight VideoLLaMA combined with

**Table 3: VBench evaluation results per dimension.**

| Models                 | Subject Consistency | Background Consistency | Temporal Flickering | Motion Smoothness    | Dynamic Degree | Aesthetic Quality | Imaging Quality | Object Class        |
|------------------------|---------------------|------------------------|---------------------|----------------------|----------------|-------------------|-----------------|---------------------|
| WAN-T2V-1.3B-480p [25] | 96.11%              | 98.06%                 | 99.09%              | 98.75%               | 27.77%         | 65.83%            | 68.91%          | 66.66%              |
| WAN-T2V-1.3B-1K [25]   | 95.86%              | 98.15%                 | 98.07%              | 98.75%               | 66.66%         | 54.82%            | 55.12%          | 33.33%              |
| UltraWAN-1K (Full)     | 95.71%              | 97.94%                 | 98.86%              | 99.06%               | 22.22%         | 61.52%            | 67.39%          | 66.66%              |
| UltraWAN-1K (LoRA)     | 97.27%              | 98.26%                 | 99.33%              | 98.62%               | 66.66%         | 62.5%             | 67.74%          | 82.29%              |
| UltraWAN-4K (LoRA)     | 96.05%              | 98.02%                 | 98.88%              | 98.47%*              | 66.66%*        | 56.81%            | 71.61%          | 50.00%              |
| Models                 | Multiple Objects    | Human Action           | Color               | Spatial Relationship | Scene          | Appearance Style  | Temporal Style  | Overall Consistency |
| WAN-T2V-1.3B-480p [25] | 51.04%              | 66.66%                 | 100.0%              | 100.0%               | 08.33%         | 20.54%            | 24.39%          | 25.31%              |
| WAN-T2V-1.3B-1K [25]   | 25.00%              | 22.22%                 | 100.0%              | 36.66%               | 00.00%         | 18.75%            | 12.24%          | 20.65%              |
| UltraWAN-1K (Full)     | 47.91%              | 66.66%                 | 100.0%              | 50.00%               | 16.66%         | 17.85%            | 19.81%          | 24.27%              |
| UltraWAN-1K (LoRA)     | 49.58%              | 66.66%                 | 100.0%              | 75.76%               | 18.22%         | 19.57%            | 23.34%          | 23.99%              |
| UltraWAN-4K (LoRA)     | 42.75%              | 66.66%                 | 100.0%              | 100.0%               | 00.00%         | 19.46%            | 19.31%          | 22.88%              |

**Figure 4: Qualitative comparison on HiVG benchmark.**

our AdaVideoRAG can be comparable to GPT-4o in long-video understanding, verifying that our AdaVideoRAG can generalize and enhance the reasoning ability of small models.

In the HiVU benchmark test, experiments show as Tab. 2: the model integrated with AdaVideoRAG has significant advantages over the original baseline, and the advantages expand with the increase of reasoning difficulty, verifying the effectiveness of its complex reasoning and the rationality of the difficulty division.

**Videos Generation of REVIG** We conduct the I2V experiment of video generation in the HiVG benchmark (only input one image), and we use the latest Wan2.1-14B as our baseline model. It can be concluded from Fig. 4 that optimizing the prompt with the retrieved "dance" action caption and "entity" description can make the generative model achieve a more matching effect.

**VBench Results of UltraWAN** Tab. 3 demonstrates that our UltraWAN [31] outperforms the baseline methods in terms of quantitative metrics, validating the effectiveness of the UltraVideo dataset [31].

## 7 Conclusion

This paper systematically introduces our works on RAG-based visual understanding and generation, centered around RAG. Concurrently, we present a high-quality UltraVideo dataset and the HiVU/HiVG benchmarks to evaluate the effectiveness of these methods. Looking ahead, we plan to extend RAG to Agent RAG with greater practical applications and continue to delve into unified visual understanding and generation research oriented toward AGI. **Legal, ethical and privacy implications.** Even though our datasets and benchmarks strive to balance as many diverse themes as possible, the models themselves can still produce hallucinatory outputs. Additionally, generated video data may be misused, so we will attach recommendations for safe usage scopes when open-sourcing the models. Legislation related to deepfake detection and generated content also needs further strengthening.

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